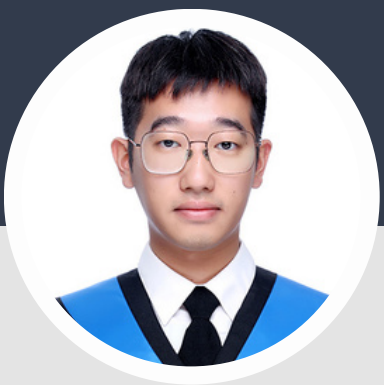
SHAO-MING RUAN

ELECTRICAL ENGINEERING STUDENT | NATIONAL CHENG KUNG UNIVERSITY



CONTACT

+886 988 106-253

bend06b@gmail.com

Tainan, Taiwan

Github

LinkedIn

My Website

SKILLS

- ML/CV
  - PyTorch
  - HuggingFace
  - OpenCV
  - Model training & evaluation
  - Inference & deployment
- Model Efficiency / Optimization
  - N:M structured pruning (2:4)
  - LoRA fine-tuning
- Engineering Tools
  - Git
  - Docker
  - Linux

CERTIFICATIONS

- Coursera – Deep Learning Specialization
- TOEIC – Score: 835
- Hackathon Finalist – Sparkful 2025
- Awarded "Most Popular Project" at 2024 AI Governance Hackathon – Tainan Digital City

PROFILE

I am currently pursuing an M.S. in the Department of Electrical Engineering at National Cheng Kung University, specializing in the Information Engineering track . During my studies, I have actively engaged in AI applications and system implementation, gaining hands-on experience in machine learning, generative AI, model compression, and full-stack integration. I have independently developed multiple prototype systems and served as a technical lead in several hackathons, with end-to-end experience from data processing and model training to real-world deployment. I am skilled at turning technology into practical, deployable solutions and enjoy collaborating with cross-functional partners to drive product validation.

EDUCATION

Bachelor of Science in Electrical Engineering 2021- 2025
National Cheng Kung University (NCKU)

Master of Science in Electrical Engineering 2025- 2027
National Cheng Kung University (NCKU)
Computer Engineering Track

PROJECTS

YOLOv8 Food Classification & Community Platform | Course Project / Competition

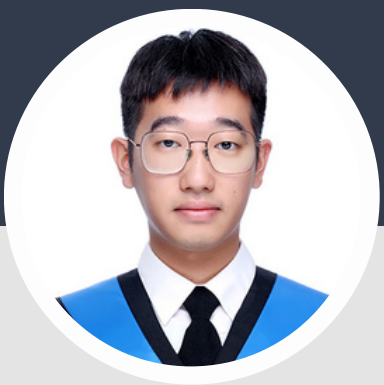
- Organized and trained on the UECFOOD256 dataset using YOLOv8 to build a food recognition model
- Built an inference and demo workflow to support real-time recognition feedback after users upload images
- Connected recognition results to platform features, extending the application to health/nutrition use cases
- Combined social interaction and challenge features; recognized as a finalist team, demonstrating commercialization potential

Whisper Auto-Subtitle Tool (Local Inference) | Tool Development

- Developed a full-stack financial dashboard integrating real-time market analysis tools.
- Developed stock evaluation strategies using Selenium + Pandas; visualized results with Highcharts and deployed system with Docker Compose.
- Planned extension: integrate LLMs or time-series forecasting models for smart investment recommendation.

DeiT Transformer Model Pruning & Lightweighting | Personal Research Project

- Applied N:M structured pruning to DeiT-Tiny and built a complete pruning → fine-tuning → evaluation workflow
- Introduced LoRA fine-tuning to reduce accuracy degradation and improve deployability
- Produced reproducible code and consolidated results



阮紹銘

國立成功大學 電機工程學系 碩士班(資訊工程組/電機乙組)

聯絡方式

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技術技能

- ML/CV：
  - PyTorch
  - HuggingFace
  - OpenCV
  - 模型訓練/評估
  - 推論與部署流程
- 模型效率化：
  - N:M 結構化剪枝（2:4）
  - LoRA 微調
- 工程工具：
  - Git
  - Docker
  - Linux

證照與競賽經歷

- Coursera 深度學習專項課程
- TOEIC 英語聽讀測驗：835 分
- Sparkful Hackathon 2025｜決賽入圍團隊
- 台南 AI 治理黑客松 2024｜榮獲「最佳人氣獎」
- 參加花蓮黑客松

個人簡介

目前就讀於國立成功大學電機工程學系，碩士班主修資訊工程組（電機己組）。在學期間積極投入 AI 技術應用與系統實作，累積了豐富的機器學習、生成式 AI、模型壓縮與前後端整合經驗。曾獨立開發多項原型系統，並於多場黑客松中擔任技術主導，熟悉從資料處理、模型訓練到實際部署的完整流程。擅長將技術轉化為可落地的解決方案，並樂於與跨領域夥伴合作推進產品驗證。

教育背景

國立成功大學 電機工程學系
預計畢業時間：2025 年 6 月

國立成功大學 電機工程學系 資訊工程組（電機己組）碩士班
預計畢業時間：2027 年 6 月

專案經歷

YOLOv8 食物分類與社群平台｜專題/競賽

- 使用 YOLOv8 進行 UECFOOD256 類別整理與訓練，完成食物辨識模型
- 建立推論與展示流程，支援使用者上傳圖片後即時辨識與回饋
- 以平台功能串接辨識結果，延伸健康/營養分析的應用情境
- 結合社交互動與挑戰功能，獲選為百組團隊中的決賽入圍者，並獲評審肯定具商業潛力。

Whisper 自動字幕工具（本地推論）｜工具開發

- 串接 yt-dlp＋FFmpeg 建立影音下載與轉檔流程，整合 Whisper 推論產生字幕
- 以 GUI 方式提供一般使用者操作，並以多執行緒避免介面卡頓
- 支援 CUDA 推論設定（依環境可切換）
- 規劃後續加入 LLM 與時序預測模組以強化智慧推薦功能。

DeiT Transformer 模型剪枝與輕量化｜個人研究專案

- 對 DeiT-Tiny 進行 N:M 結構化剪枝（2:4、4:8），建立剪枝→微調→評估的實作流程
- 導入 LoRA 進行微調，以降低精度損失並提升可部署性
- 產出可重複實驗的程式流程與結果整理（可在 GitHub 提供佐證）

AI 財務儀表板系統｜Django 全端專案

- 使用 Django 建置金融儀表板，整合資料查詢與互動式視覺化
- 以 Selenium + Pandas 建立資料蒐集與清理流程，提升更新效率與一致性
- 規劃擴充：加入模型化分析/預測模組與更多指標報表